PRACTICALS

Computer Networks Laboratory

Practical 2.

Introduction to Networking Terminal Commands

1. Introduction

A computer network is built from end devices, switches and routers, and each of them can be examined and controlled through a text-based terminal. This command-line interface (CLI) is the most direct way to see how a network behaves and to change it.

Some commands are run at the command prompt of an ordinary computer to check addressing and test connectivity, such as ipconfig and ping. Others are entered on Cisco devices running IOS to name the device, address its interfaces, define routes and verify the result. This document presents 27 essential commands, each listed once, in the order in which they are usually used.

2. Understanding the Command-Line Environment

Cisco IOS modes

IOS protects the device by dividing the command set into modes. Each mode has its own prompt and allows only certain commands. A user starts in the least powerful mode and moves upward as more control is needed. Recognising the prompt tells the administrator which mode is active.

Mode	Prompt	Reached by	Purpose
User EXEC	`Router>`	`Logging in`	Basic checks with limited commands
Privileged EXEC	`Router#`	`enable`	All show commands and saving the configuration
Global configuration	`Router(config)#`	`configure terminal`	Device-wide settings such as the hostname
Interface configuration	`Router(config-if)#`	`interface <type/number>`	Settings for one or more interfaces
Line configuration	`Router(config-line)#`	`line console 0`	Settings for console access
Router configuration	`Router(config-router)#`	`router rip`	Settings for a routing protocol

Table 1: Cisco IOS command modes

Reading the command tables

In the tables that follow, text written in angle brackets, such as <IP> or <network>, is a placeholder to be replaced with a real value. Words in plain text are typed exactly as shown. The commands are grouped by category, numbered 1 to 27, and appear in the order in which they are normally used.

Category	Commands	Main use
PC Prompt	4	Check host addressing and test reachability
Mode Navigation	4	Move between IOS modes
Device Setup	4	Name the device and secure access
Interface Config	3	Address and activate interfaces
Switch Only	2	Enable remote management of a switch
Static Routing	2	Define routes manually
RIP	4	Share routes dynamically between routers
Verification	3	Confirm configuration and operation
Save	1	Store the configuration permanently

Table 1a: Overview of command categories

3. Commands Used at the PC Command Prompt

These commands are typed at the command prompt of an end device such as a PC or laptop. They are used to examine the addressing of the host and to test whether it can reach other devices and services.

#	Command	Purpose	Explanation
1	`ipconfig`	Show IP, mask, gateway	Displays the IPv4 address, subnet mask and default gateway of every network adapter. Usually the first command run when checking a host.
2	`ping <IP>`	Test connectivity	Sends ICMP echo requests and reports replies, round-trip time and packet loss. A reply confirms that the destination is reachable.
3	`tracert <IP>`	Show hops to destination	Lists each router (hop) on the path with its response time, which helps locate the point where a path fails.
4	`nslookup <name>`	DNS lookup	Queries the DNS server to translate a domain name into an IP address and confirms that name resolution works.

Table 2: PC command prompt commands

4. Mode Navigation Commands

Cisco IOS is organised into modes, and a command works only in the mode meant for it. The commands below are used to move between modes (see Table 1).

#	Command	Purpose	Explanation
5	`enable`	Go to privileged mode	Moves from user EXEC mode (>) to privileged EXEC mode (#). A password may be requested if one is configured.
6	`configure terminal`	Go to global config mode	Enters global configuration mode, where changes to the running configuration begin.
7	`interface <type/number>`	Enter interface mode	Selects a single interface, such as FastEthernet, GigabitEthernet or Serial, for configuration.
8	`exit`	Go back one level	Returns to the previous mode, for example from interface mode to global configuration mode.

Table 3: Mode navigation commands

5. Device Setup Commands

These commands give a device its identity and protect access to it. They are normally the first configuration entered on a new router or switch.

#	Command	Purpose	Explanation
9	`hostname R1`	Set device name	Changes the device name shown in the prompt, making it easy to tell devices apart.
10	`enable secret <password>`	Encrypted privileged password	Protects privileged mode with a password stored in hashed form. It is preferred over the older enable password command.
11	`line console 0`	Enter console line config	Enters configuration mode for the console port, the direct cable connection to the device.
12	`password <pw> and login`	Secure console access	Two commands used together in line mode: password sets the console password and login makes the device ask for it.

Table 4: Device setup commands

6. Interface Configuration Commands

Interfaces are the points where a device connects to a network. They must be addressed and switched on before they can carry traffic.

#	Command	Purpose	Explanation
13	`ip address <IP> <mask>`	Assign IP and subnet mask	Gives the interface its IPv4 address and subnet mask, placing it in a network.
14	`no shutdown`	Turn interface on	Enables the interface. Router interfaces are administratively down by default and stay inactive until this is entered.
15	`clock rate 64000`	Set clock on DCE end of serial link	Supplies the timing signal on the DCE end of a serial link. The DTE end receives its clock from this side.

Table 5: Interface configuration commands

7. Switch-Only Commands

A Layer 2 switch forwards frames by MAC address and does not need an IP address for that task. An address is needed only so that the switch itself can be managed remotely.

#	Command	Purpose	Explanation
16	`interface vlan 1`	Management interface of switch	Enters the virtual interface used to manage the switch. It needs an IP address and no shutdown to become active.
17	`ip default-gateway <IP>`	Gateway for the switch	Tells the switch which router to use for management traffic destined for other networks.

Table 6: Switch management commands

8. Static Routing Commands

A router can forward packets only to networks it knows about. With static routing, the administrator enters these routes manually. Static routes are simple, predictable and use no bandwidth for updates, but they do not adapt to changes by themselves.

#	Command	Purpose	Explanation
18	`ip route <network> <mask> <next-hop>`	Add static route	Defines a path to a remote network through the IP address of the next router.
19	`ip route 0.0.0.0 0.0.0.0 <next-hop>`	Default route	Matches any destination not found in the routing table, often used towards the Internet or a single upstream router.

Table 7: Static routing commands

9. RIP (Dynamic Routing) Commands

Routing Information Protocol (RIP) is a distance-vector protocol that lets routers share routes automatically and uses hop count as its metric, with a maximum of 15 hops. Routers learn of network changes without manual editing, which makes RIP a suitable first dynamic protocol for beginners.

#	Command	Purpose	Explanation
20	`router rip`	Enable RIP	Starts the RIP process and enters router configuration mode.
21	`version 2`	Use RIPv2	Selects version 2, which carries subnet masks and supports VLSM and CIDR, unlike version 1.
22	`network <network-address>`	Advertise connected network	Names a directly connected network. RIP then runs on matching interfaces and advertises that network.
23	`no auto-summary`	Disable auto-summarization	Stops RIP from summarising routes to classful boundaries, which can otherwise cause incorrect routing between subnets.

Table 8: RIP configuration commands

10. Verification Commands

After configuring a device, show commands confirm that the settings are correct and the device is behaving as expected. They are entered in privileged EXEC mode and do not change the configuration.

#	Command	Purpose	Explanation
24	`show running-config`	Current configuration	Displays the configuration currently active in RAM, including changes not yet saved.
25	`show ip interface brief`	Interface status and IPs	Gives a one-line summary of each interface with its IP address and status. Often the quickest way to spot a disabled or unaddressed port.
26	`show ip route`	Routing table	Lists the networks the router knows about and how each was learned, such as connected, static or RIP.

Table 9: Verification commands

11. Saving the Configuration

Changes made in configuration mode take effect immediately, but they are held only in the running configuration in RAM. Unless they are saved, they are lost when the device loses power or reloads.

#	Command	Purpose	Explanation
27	`copy running-config startup-config`	Save configuration	Copies the running configuration to NVRAM so that it is used the next time the device starts.

Table 10: Save command

12. The Logical Order of Using Commands

The command list follows the order in which an administrator normally works. Understanding this sequence matters as much as knowing the individual commands.

1. **Check the host.** PC prompt commands confirm the addressing and basic reachability of end devices.
2. **Enter the right mode.** Mode navigation commands move from user mode to global configuration and then to the interface or line to be configured.
3. **Set up the device.** The device is named and its console and privileged access are secured.
4. **Configure interfaces.** Addresses are assigned, interfaces are enabled, and clocking is set where required. A switch is given a management address.
5. **Configure routing.** Static routes, a default route or RIP are added so that networks can reach one another.
6. **Verify.** Show commands confirm that interfaces, routes and protocols match the intended design.
7. **Save.** The running configuration is copied to the startup configuration.

13. Good Practices and Common Mistakes

Good practices

- **Check the prompt.** Always confirm the current mode before typing a command.
- **Verify after configuring.** Use show commands to confirm that each change had the intended effect.
- **Protect access.** Set an enable secret and a console password on every device.
- **Save deliberately.** Copy the running configuration to the startup configuration once the changes are confirmed.

- **Work from the basics.** Test the host first with ipconfig and ping, then move on to the device configuration.

Common mistakes

- **Forgetting no shutdown.** An interface with a correct address still passes no traffic while it is administratively down.
- **Not saving the configuration.** All changes disappear when the device restarts.
- **Wrong network statement.** RIP advertises only the networks entered, so a missing or incorrect entry leaves that network unreachable.
- **Missing clock rate.** A serial link does not come up if no DCE end supplies the clock.
- **No gateway on a switch.** The switch cannot be managed from another network without a default gateway.

Conclusion

Networking terminal commands are the working language of network administration. The 27 commands covered here begin with simple tools on a host, move through the IOS modes, device setup, interface and routing configuration, and end with verification and saving. Mastering them in this order builds the habit of configuring carefully, checking the result and preserving it. With regular practice, these basic commands form a strong foundation for more advanced topics such as VLANs, OSPF, access control lists and network security.